

MULTIMEDIA



UNIVERSITY

STUDENT ID NO

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MULTIMEDIA UNIVERSITY

FINAL EXAMINATION

TRIMESTER OCTOBER/NOVEMBER 2025 (TERM 2530)

GBB6263 – MANAGEMENT DECISION SCIENCE

(All sections / Groups)

3 MARCH 2026

9.00 a.m. – 11.00 a.m.

(2 Hours)

INSTRUCTIONS TO STUDENT

1. This question paper consists of **FOUR (4)** printed pages with 4 questions only, excluding the cover page.
2. Attempt **ALL FOUR (4)** questions.
3. Write all your answers in the answer booklet provided. All necessary workings **MUST** be shown.

Question 1 (25 marks)

Read Bookstore publishes three types of books X, Y and Z. The selling price per book is RM5 for book X, RM6 for book Y and RM7 for book Z. The time required to sell one book is 8 minutes for book X, 10 minutes for book Y and 12 minutes for book Z. The delivery cost per book is RM1.50 for book X, RM2 for book Y and RM1 for book Z. The maximum available selling time per week is 1500 minutes and the maximum delivery expenses per week is RM120. The company wishes to maximize its total weekly revenue.

- (a) Define the decision variables. [3 marks]
- (b) Write the objective function to maximize its total weekly revenue. [2 marks]
- (c) Formulate the constraints of the problem. [3 marks]
- (d) Convert the problem into standard form and construct the Initial Simplex Tableau. [7 marks]

The final simplex tableau for this problem is given below:

C_j		5	6	7	0	0	RHS
	Solution Mix	x	y	z	s_1	s_2	Quantity
0	s_1	-10	-14	0	1	-12	60
7	z	1.5	2	1	0	1	120
	Z_j	10.5	14	7	0	7	840
	$C_j - Z_j$	5.5	8	0	0	7	

s_1 = slack variable for selling time (minutes)

s_2 = slack variable for delivery expenses

- (e) Use the final simplex tableau to determine the optimal values of X, Y and Z. Next, state the maximum weekly profit. [5 marks]
- (f) Which of the constraints are binding and which are non-binding? [3 marks]
- (g) Identify the resource that is not fully utilized and state the unused amount. [2 marks]
- Continued...**

Question 2 (25 marks)

- (a) A consumer electronics company manufactures Smart TVs at four production plants and distributes them to three retail outlets in West Malaysia. The transportation cost per unit from each plant to each outlet, together with the available supply at each plant and the demand at each outlet is shown in the table below:

Production Plant	Outlet			Supply (units)
	A	B	C	
Penang	45	60	55	250
Ipoh	50	48	52	300
Shah Alam	65	70	60	350
Kuantan	58	53	57	300
Demand (units)	400	450	350	

Using the Northwest Corner Method, determine an initial feasible solution for the transportation problem.

[6 marks]

- (b) A consulting firm plans to assign five consultants to five regional offices, namely Kuala Lumpur, Penang, Kuching, Kota Kinabalu and Johor Bahru. Each consultant can be assigned to only one office, and each office requires exactly one consultant. The estimated assignment cost (in RM) for each consultant office combination is given as follows:

Consultant	Regional office				
	Kuala Lumpur	Penang	Kuching	Kota Kinabalu	Johor Bahru
Amir	82	83	69	92	87
Brian	77	37	49	92	88
Clara	11	69	50	86	92
Kate	8	9	98	23	32
Siva	31	83	73	63	45

Apply the Hungarian Method to determine the optimal assignment plans and state the total minimum assignment cost.

[19 marks]

Continued...

Question 3 (25 marks)

Vertex Innovations Sdn. Bhd. has been contracted to implement a university digital library system. The project involves several interdependent activities that must be carefully planned to avoid delays. The project manager has identified ten major activities (A to J), along with their immediate predecessors and estimated completion times (in weeks). To ensure successful delivery, you are required to analyze the project using the Critical Path Method (CPM).

Activity	Immediate Predecessors	Activity Time
A	-	3
B	-	5
C	A	4
D	A	6
E	B	3
F	C	5
G	D, E	4
H	F	2
I	G	3
J	H, I	4

- (a) Draw a network diagram that clearly shows all activities and their precedence relationships.

[6 marks]

- (b) Using the network diagram, calculate the Earliest Start (ES), Earliest Finish (EF), Latest Start (LS), Latest Finish (LF), Slack for each activity, then complete the table in your answer booklet.

Activity	ES	EF	LS	LF	Slack	Critical Path?
A						
B						
C						
D						
E						
F						
G						
H						
I						
J						

[12 marks]

Continued...

(c) Identify the critical path of the project.

[5 marks]

(d) Determine the expected project completion time.

[2 marks]

Question 4 (25 marks)

A company is considering expanding its smartphone assembly facility to meet anticipated market demand. They need to decide whether to expand to a Small, Medium or Large facility. The company has prepared the following payoff table (profits in RM) and estimated probabilities for different market conditions:

Size	Demand (RM)		
	High	Medium	Low
Small	180 000	120 000	-50 000
Medium	400 000	250 000	-100 000
Large	700 000	300 000	-250 000
Probability	0.3	0.5	0.2

Determine the best expansion decision using the following decision criteria:

(a) Maximax.

[3 marks]

(b) Maximin.

[3 marks]

(c) Equally Likely Criterion (Laplace Criterion).

[4 marks]

(d) Criterion of Realism (Hurwicz), $\alpha = 0.7$

[4 marks]

(e) Expected Monetary Value (EMV).

[4 marks]

(f) Minimax Regret Criterion.

[4 marks]

(g) Value of Perfect Information.

[3 marks]

End of Page.

